MEDBUDDY: AI-Powered Medical Chatbot with Retrieval-Augmented Generation

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Guide

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The Evolution of Security



- Abstract & Introduction
- Literature Survey
- Problem Identification
- Existing System & Disadvantages (The Monolithic Cloud)
- Proposed System & Our Contribution (The Hybrid Blueprint)
- System Implementation & Architecture
- Results, Conclusion & Future Scope

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Executive Summary: The Engineering Pivot

- **The Risk:** Generic AI lacks source transparency and fabricates syllabus data.
- **The Transition:** Shifted from rate-limited `Gemini APIs` to the high-speed `Grok LLM` to eliminate API exhaustion.
- **The Engineering:** Integrated `Local FastEmbed` for on-device vectorization and persistent `MD5 hashing` for fault-tolerant processing.
- **The Validation:** Implemented the `TruLens RAG Triad` for real-time quality scoring.

84%

Response accuracy rate for single medical textbooks.

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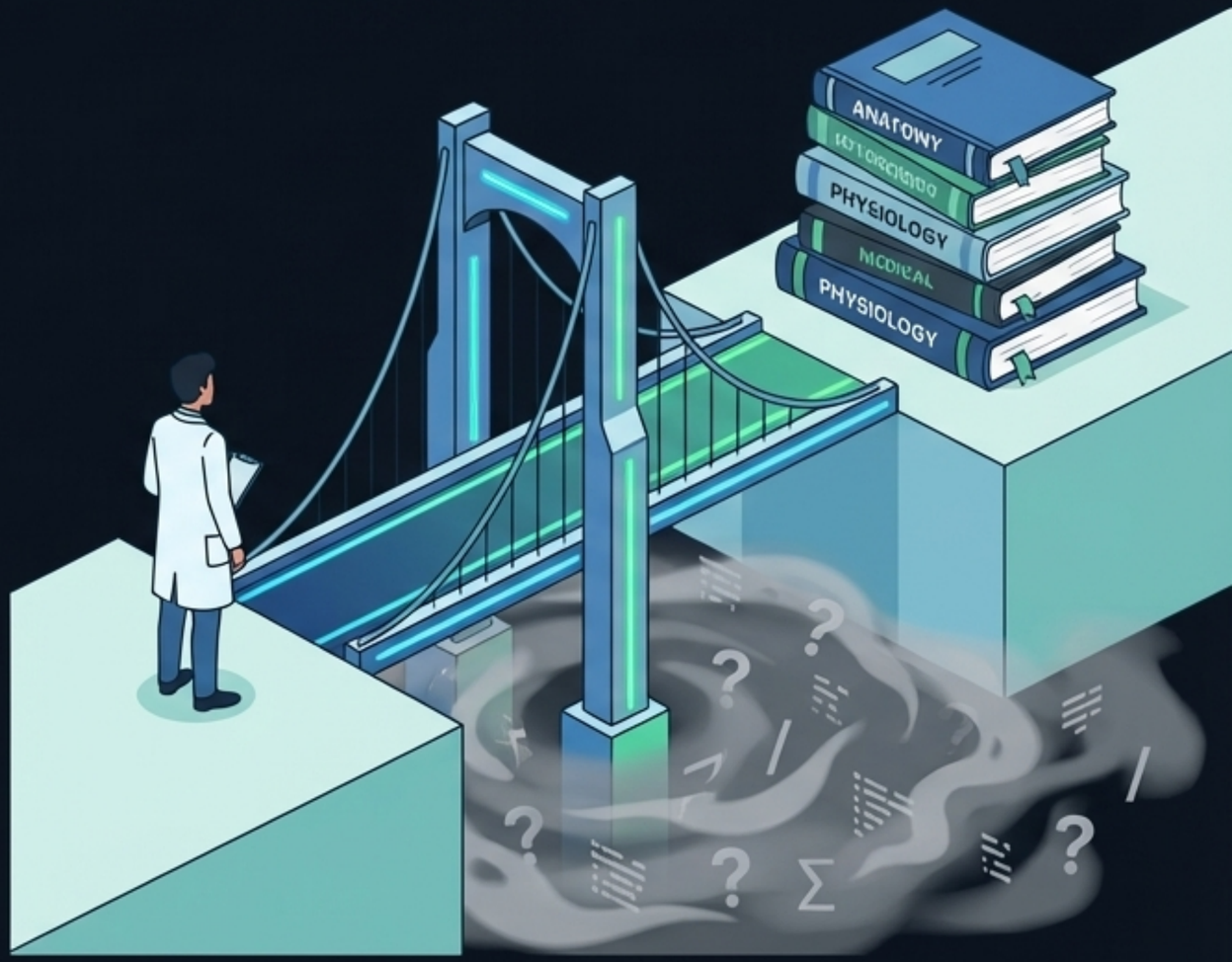
Bridging the Medical Education Gap

Rising demand for skilled healthcare professionals necessitates better, faster medical preparation tools.

Generic AI models lack transparency and carry a fatally high risk of hallucination.

MedBuddy serves as a digital assistant tailored to the user's exact focus area.

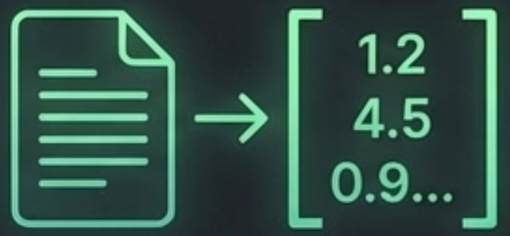
The goal: Shift the application focus from slow data retrieval to true medical comprehension.



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Three Pillars of Safe Medical AI

Embeddings



Turning human sentences into mathematical number-lists so the computer understands exact medical context.

Vector Database



A specialized localized storage space for instant mathematical matching of user questions to verified textbook paragraphs.

LLM Grounding



A strict software rule forcing the AI to only generate answers using the uploaded text, completely stopping outside fabrication.

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Modern Technology Stack

Streamlit

Web-based interface and session management.

LlamaIndex & FastEmbed

Core data framework and local, efficient text embeddings.

Grok & Gemini

High-performance LLM providers for response generation.

TruLens

Real-time evaluation of AI groundedness.

PyMuPDFReader

Specialized PDF extraction for dense medical text (supports up to 200MB).

Hashlib & LiteLLM

MD5 persistent index caching and universal adapter components.

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Literature Survey: The Promise and the Peril

Accelerated Potential

Investigates standard LLMs in medical learning and vaccination training (Baglivo et al.).

Demonstrates that AI tools effectively accelerate medical research cycles.

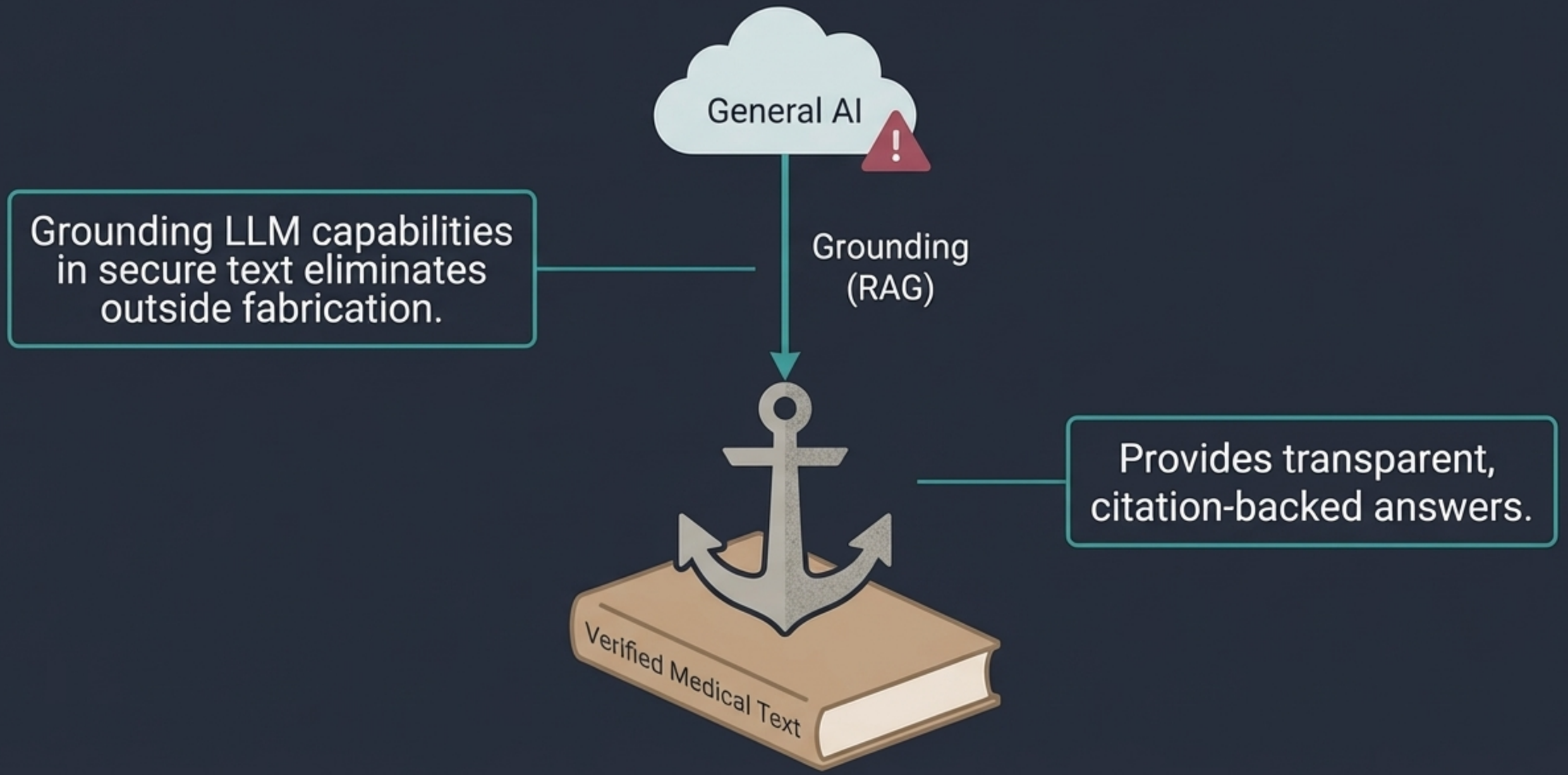
Critical Barriers

Uncovers a high risk of authenticity issues without a dedicated retrieval system.

Concludes that professional verification and reliable source-tracking are mandatory for safety.

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The Hallucination Anchor



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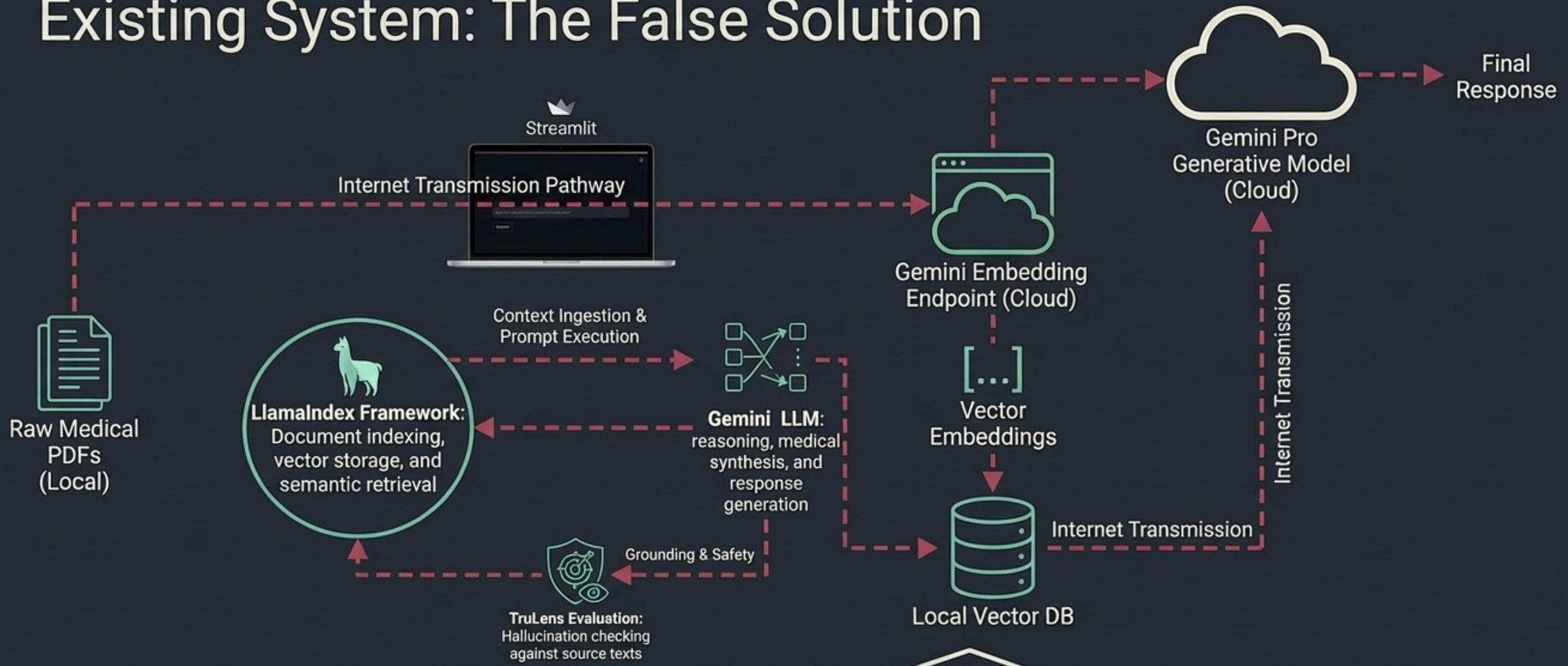
The Goliath: System Vulnerabilities

- **Lack of Transparency:** Generic AI is not concrete about the source of its information.
- **The Hallucination Risk:** Standard LLMs have a high possibility of fabricating facts.
- **Vulnerability:** Students are exposed to false information or data unrelated to their syllabus.
- **Critical Consequences:** Misleading medical students with wrong information might cost lives.



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Existing System: The False Solution



Monolithic Cloud Architecture: Relies entirely on the external API ecosystem for every NLP pipeline stage.

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Threat Matrix: The Cost of Cloud Dependency

Severe Data Privacy Risks

Uploading proprietary medical documents and sensitive patient data to third-party servers inherently violates strict medical data privacy laws.

Exorbitant API Costs

Generating embeddings for thousands of pages of medical textbooks consumes millions of tokens, leading to unsustainable expenses.

High Latency

Entirely bottlenecked by internet bandwidth. Uploading large documents for cloud vectorization is painfully slow.

Rate Limiting

Cloud providers impose strict Requests/Tokens Per Minute limits. Ingesting a large medical corpus results in API crashes and timeouts.

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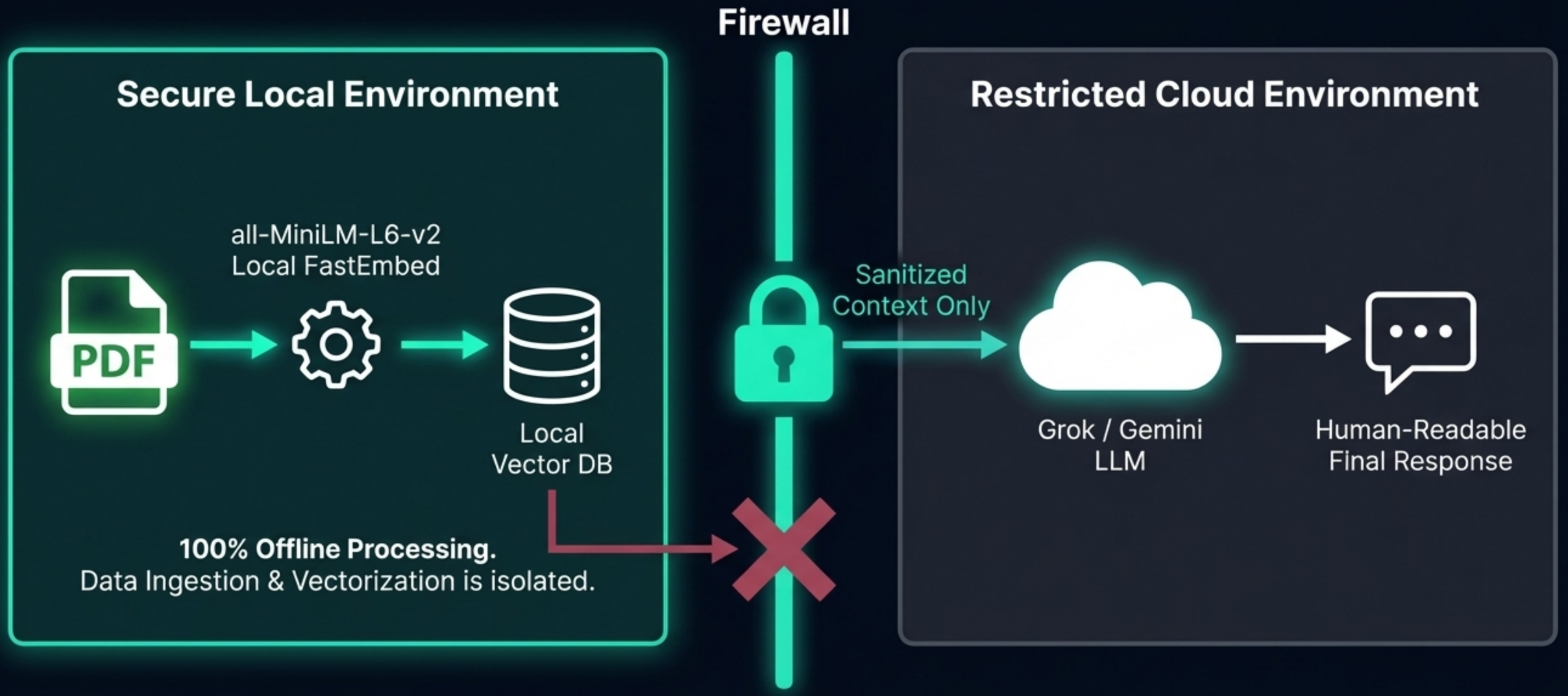
The True Solution: MedBuddy Hybrid Framework



- Introduces a sophisticated, hybrid RAG framework tailored for healthcare environments.
- Bifurcates the workload to maximize both security and artificial intelligence capabilities.
- Processes, cleans, chunks, and vectorizes all medical data locally on the host machine.
- Explicitly restricts the cloud-based LLM from accessing the raw data ingestion phase.

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The Bifurcated Workload



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Architectural Shift: Local Embeddings

- Contribution: Hugging Face Local Embedding Integration.
- The Tech: Deployed the SentenceTransformers library, specifically the lightweight, mathematically efficient all-MiniLM-L6-v2 model.
- The Mechanism: Maps sentences to a 384-dimensional dense vector space entirely offline, optimized for semantic similarity.
- The Impact: Bypasses cloud rate limits, utilizes host CPU/GPU, and reduces overall cloud token consumption by over 50%.

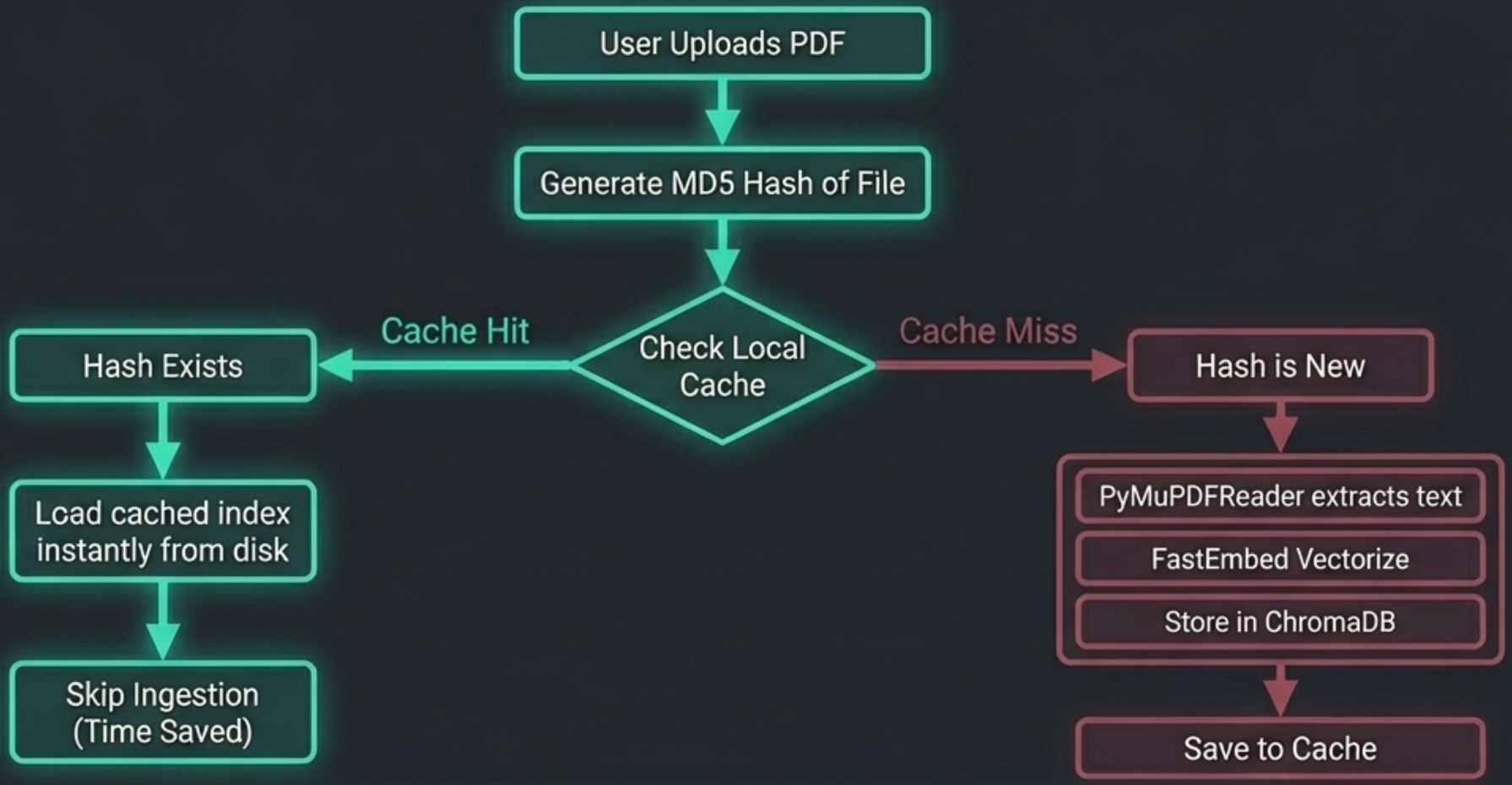
```
from fastembed import TextEmbedding

# Replaces Google API call
embedding_model = TextEmbedding(model_name
='BAAI/bge-small-en-v1.5')

documents = SimpleDirectoryReader('./data'
).load_data()
vector_index = VectorStoreIndex.from_
documents(
    documents, embed_model=embedding_model
)
```

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Fault-Tolerant Indexing: MD5 Checkpoint



This architecture creates a fault-tolerant mechanism allowing interrupted document processing to resume seamlessly.

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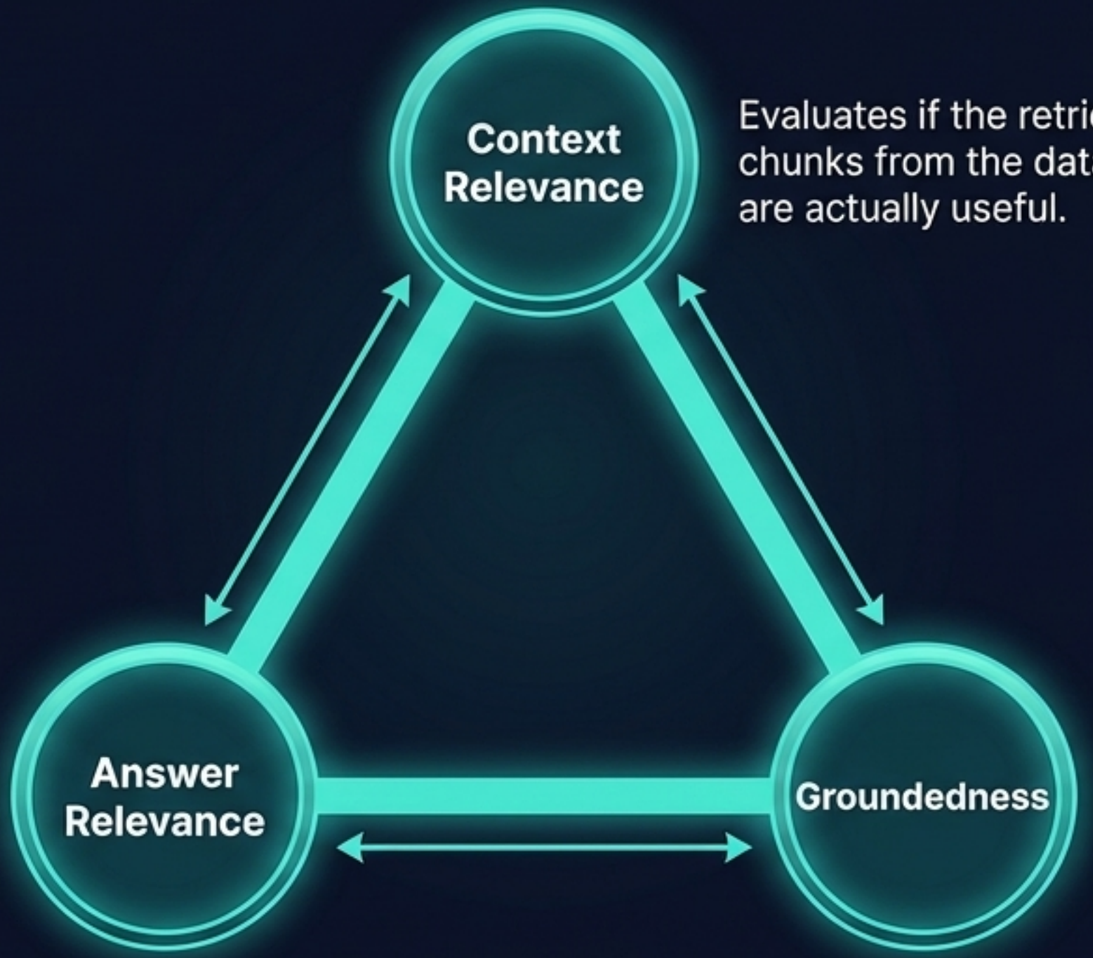
Monolithic vs. Hybrid Architecture

	Existing Base System	Proposed MedBuddy System
Document Ingestion Location	Cloud (Vulnerable)	Local Host Machine (Secure)
Embedding Token Cost	High / Escalating	Absolute Zero
Privacy Standard	Non-compliant / Exposes Data	High Security / Offline
API Exhaustion Risk	High (Rate limited)	Eliminated (Local processing)

The hybrid approach yields massive systemic improvements without sacrificing generative intelligence.

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The TruLens RAG Triad



Evaluates if the retrieved chunks from the database are actually useful.

Ensures the final generated answer directly addresses the original question.

Checks the LLM's final response against the retrieved text to ensure zero fabrication.

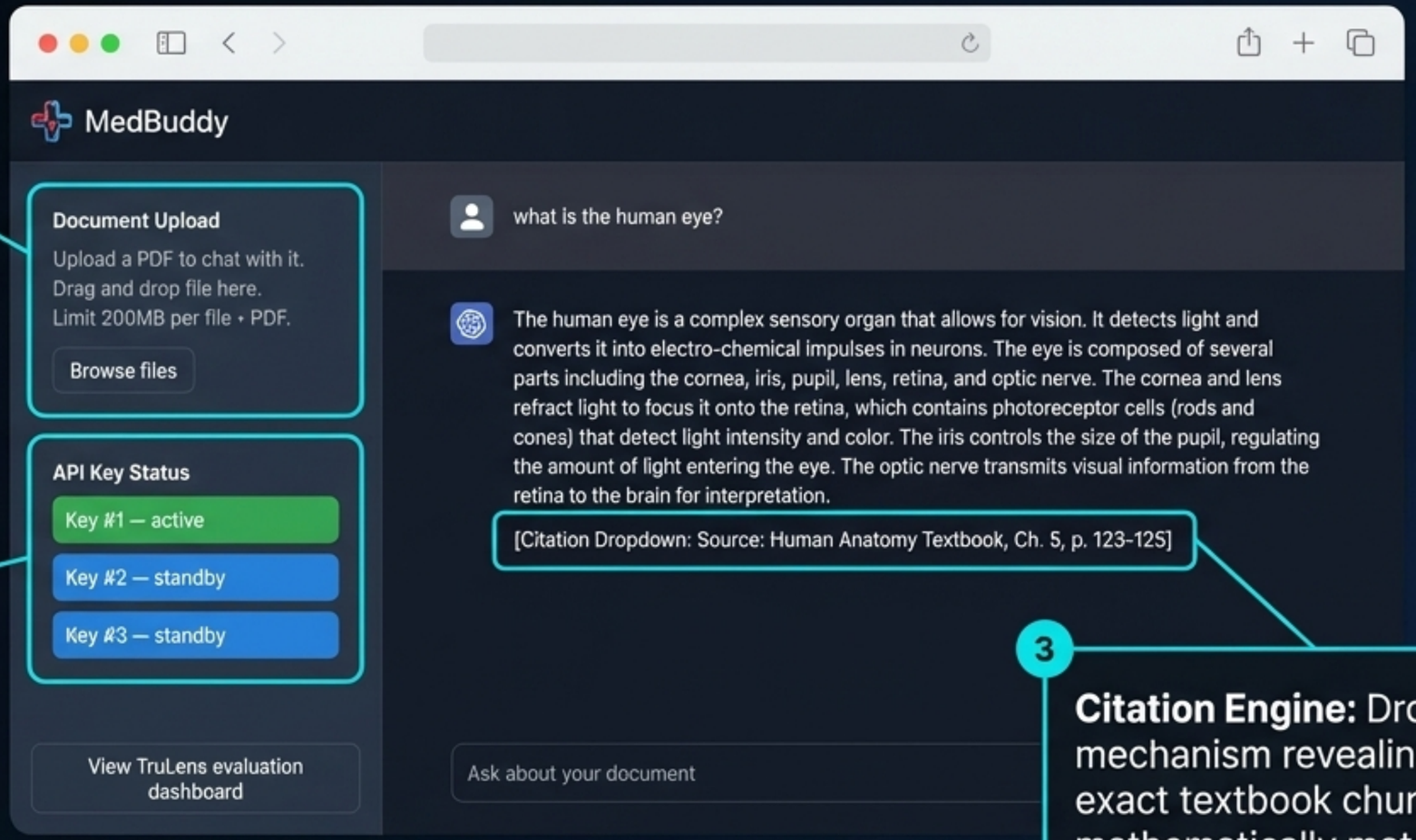
Scores are returned to the user interface in real-time giving immediate, mathematically verified feedback on response reliability.

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System Implementation

1 **Secure Upload:** Frictionless interface supporting dense PDFs up to 200MB entirely locally.

2 **Cryptographic API Rotation:** API key status panel showing standby keys and handling quota failures seamlessly.



3 **Citation Engine:** Dropdown mechanism revealing the exact textbook chunks mathematically matched to the query.

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Quantitative Validation



Ingestion Speed

Local Hugging Face models processed large medical PDFs significantly faster by eliminating HTTP request round-trips.

$0

Cost Reduction

API token usage during the document upload and vectorization phase reduced to absolute zero.



Retrieval Accuracy

The local all-MiniLM-L6-v2 model matched enterprise APIs, verifying textbook context with an 84% accuracy rate.

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Conclusion



Successfully demonstrated the **efficacy** of a **domain-specific, localized RAG** framework.

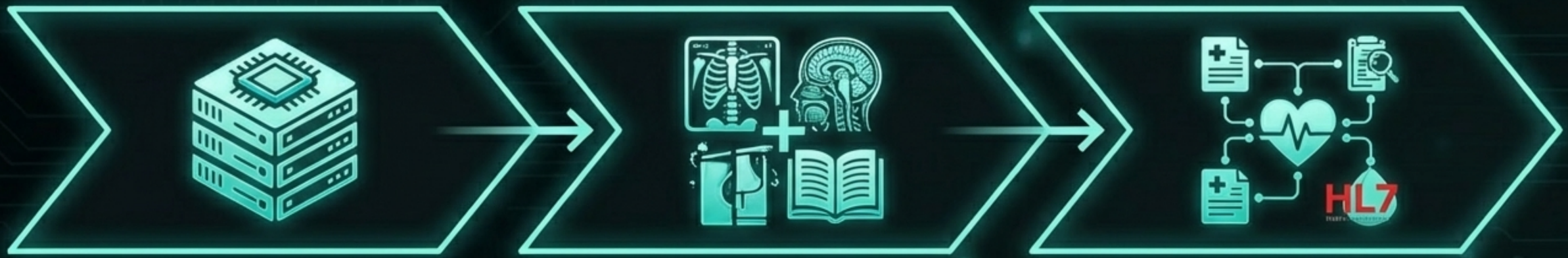
Achieved an **optimal balance** between **high-level AI reasoning** and **stringent data privacy** by integrating local open-source transformers.

Drastically mitigated risks: Eliminated external exposure during ingestion, cut token costs, and maintained factual accuracy.

MedBuddy proves to be a **viable, safe, and transparent** tool for medical students and professionals.

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Future Scope & Scalability



Fully Localized Inference

Transitioning to locally hosted LLMs (like Llama 3) to execute entirely on hospital hardware, removing the final cloud generation step.

Multi-Modal RAG

Expanding ingestion to analyze medical imagery (X-rays, MRIs, pathology reports) alongside textual textbooks for holistic diagnosis.

EHR Integration

Utilizing HL7 or FHIR standards to cross-reference a patient's real-time electronic health record against the uploaded medical guidelines.

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